KENYA MEDICAL TRAINING COLLEGE

FACULTY OF DIAGNOSTIC SCIENCES

DEPARTMENT OF MEDICAL LABORATORY SCIENCES

DIPLOMA IN MEDICAL LABORATORY SCIENCES

YEAR I, SEMESTER I EXAMINATIONS (1.1)

PAPER: CHEMISTRY

DATE:10 /02 /2025 **TIME: 2 HOURS 2.00 – 4.00 PM**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (20 Multiple choice questions)
   - Section 2 (8 short answer questions)
   - Section 3 (2 long answer questions)
2. Attempt ALL Questions
3. Write the COLLEGE NUMBER on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Any examination malpractices will be handled as per the College Examination Policy
6. Ensure you sign the examination register provided

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SECTION 1: MULTIPLE CHOICE QUESTIONS (20 MARKS)

1. A pure substance that cannot be separated into any other simple substance by chemical or physical means is an:
   A. Atom
   B. Element
   C. Compound
   D. Mixture
2. A carbon atom can form _____ covalent bonds with other atoms?
   A. 1
   B. 2
   C. 3
   D. 4
3. The smallest particle of an element that takes part in a chemical reaction is:
   A. Isotope
   B. Atom
   C. Electron
   D. Nucleus
4. Which of these is a sub-atomic particle:
   A. Alpha
   B. X-rays
   C. Protons
   D. Gamma
5. A Bronsted-Lowry acid is a:
   A. Proton donor
   B. Proton acceptor
   C. Electrons donor
   D. Hydroxyl acceptor
6. Bond formed when electrons are equally shared:
   A. Electronic
   B. Covalent
   C. Ionic
   D. Hydrogen
7. Which of these laws is explained by Dalton's atomic theory?
   A. Law of conservation of electrons
   B. Law of loss of electrons
   C. Law of definite proportions
   D. Law of multiple energy levels
8. Who used the cathode ray tube to conclude the presence of positively charged particles in an atom?
   A. John Dalton
   B. J. Thompsons
   C. James Chadwick
   D. Eugen Goldstein
9. A substance that is able to resist minor pH changes is a:
   A. Weak acid
   B. Buffer
   C. Weak base
   D. Conjugate base

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10. Calculate the pH of 1L sodium hydroxide whose concentration is 1.7×10^{-3}:
    A. 2.77
    B. 11.23
    C. 2.47
    D. 11.53
11. The difference between isotopes is:
    A. Number of electrons
    B. Atomic number
    C. Number of protons
    D. Mass number
12. Which one is the lightest:
    A. Atom
    B. Proton
    C. Electron
    D. Neutron
13. What is the correct order of reactivity?
    A. F>Cl>Br>I
    B. F>Br>Cl>I
    C. F>I>Br>Cl
    D. F>Br>I>Cl
14. Which type of chemical reaction is this; $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$?
    A. Single displacement
    B. Double displacement
    C. Synthesis
    D. Decomposition
15. Electrons in the outer energy level are referred to as:
    A. Unstable electrons
    B. Free electrons
    C. Valence electrons
    D. Protons
16. A method that can be used to separate a compound:
    A. Decantation
    B. Hydrolysis
    C. Sieving
    D. Evaporation
17. Phenolphthalein is a:
    A. Weak base
    B. Alkalized buffer
    C. Neutralized buffer
    D. Weak acid
18. The nucleus of an atom has:
    A. Delta charge
    B. Gamma charge
    C. Positive charge
    D. Negative charge
19. What is the concentration of a liter of a monoprotic acid whose pH is 1.30?
    A. 0.1
    B. 0.05
    C. 0.01
    D. 0.5

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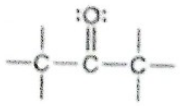
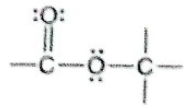
20. Alkanes are stable compounds; however, they undergo _____ reaction:
    A. Addition
    B. Substitution
    C. Dimerization
    D. Dehydration

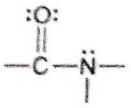
SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

21. State the five (5) differences between electrons and protons (5marks)
22. Explain the characteristics that define the solid state of matter (5marks)
23. Explain how you would separate a homogenous mixture of liquids that boils at 72 and 95 degrees Celsius respectively (5marks)
24. 20mls of sodium hydroxide solution was used to neutralize 35mls of 0.2M sulfuric acid. Calculate the concentration of the sodium hydroxide used. (Both solutions are pure) (5marks)
25. Identify the following functional groups (5marks)

i. 

ii.

iii. 

iv. 

v. $R–CH_3$

26. Calculate the empirical formula of a hydrocarbon whose combustion analysis produced 23.093g CO_2 and 4.701g H_2O. (5marks)
27. List five (5) states of matter (5marks)
28. Differentiate between primary and tertiary amines (5marks)

SECTION 3: LONG ANSWER QUESTIONS (40 MARKS)

29. Using a suitable example in each case, describe four types of chemical bonds. (20 marks)
30. Discuss four colligative properties of solutions (20 marks)

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